203739: felsic metavolcanic rock, Lake Carey

(Kurnalpi Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Laverton (SH 51-2), Laverton (3340)
MGA Zone 51, 436199E 6796702N

Sampled on 2 September 2020

This sample was collected from the 445.9 – 448.3 m depth interval of drillcore LCDD0013, drilled in 2016 by GSM Mining Company Pty Ltd at their Northern Fleet prospect. The drillhole is located on Mount Weld Station, in the northeastern part of Lake Carey, about 20 km southeast of Mount Margaret, 13.0 km north-northwest of Sunrise Dam, and 10.3 km west-southwest of Brumby Well.

Geological context

The unit sampled is a felsic metavolcanic rock (Fig. 1) within the Laverton greenstone belt of the Kurnalpi Terrane (Stewart, 2001; Standing, 2008). The unit consists of dacitic flows and breccias, within a thick metasedimentary and minor felsic metavolcanic succession in the Lake Carey domain (Standing, 2008). A porphyritic metasyenite about 0.9 km to the northeast in drillcore LCDD0076 yielded an igneous crystallization age of 2663 ± 8 Ma (GSWA 203738, Wingate et al., 2022a). A porphyritic metasyenite at the Wallaby mine, about 11.6 km to the north-northwest, indicated an igneous crystallization age of 2688 ± 27 Ma (GSWA 203740, Wingate et al., 2022b). Detrital zircons in a sandstone horizon within conglomerate at the Wallaby mine, about 11.8 km to the north-northwest, yielded a conservative maximum depositional age of 2673 ± 5 Ma (sample E319, Kositcin et al., 2008). An andesite-derived sandstone at the Sunrise Dam mine, about 15.8 km to the south-southeast, yielded a conservative maximum depositional age of 2715 ± 5 Ma (sample E324, Kositcin et al., 2008). Detrital zircons in a polymictic breccia unit at the Granny Smith mine, about 18.4 km to the north-northeast, indicated a youngest detrital age component at 2663 ± 5 Ma (sample E401, Kositcin et al., 2008). A feldspar–quartz-porphyritic rock at Ida Hill, about 27 km to the north-northeast, provided an igneous crystallization age of c. 2660 Ma (sample E258, Kositcin et al., 2008).

Petrographic description

The sample is a sericite- and carbonate-altered, autoclastic metavolcanic breccia (Fig. 1). Coarse, subangular to angular clasts are up to 1.5 cm in size (in the thin section), and consist of weakly porphyritic dacite, in which minor rounded to embayed quartz phenocrysts are dispersed within a very fine-grained, recrystallized quartzofeldspathic groundmass. The crude jigsaw fit of some clasts indicate a possible hyaloclastite. Clasts are separated by mosaics of very fine-grained quartz. Numerous stringers of sericite traverse the clasts and groundmass and define a foliation. Granular aggregates and discontinuous stringers of carbonate minerals are associated with sericite. The sample contains numerous veinlets of fine-grained granoblastic quartz. Sparse pyrite grains mantled by minor fibrous quartz fringes are closely associated with carbonate minerals.

Zircon morphology

Zircons isolated from this sample are mainly colourless and subhedral to euhedral. The crystals are up to 150 µm long, and equant to slightly elongate, with aspect ratios less than 3:1. In cathodoluminescence (CL) images, the zircons exhibit concentric and sector zoning. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed over three sessions, on 9–10, 15–16, and 22–23 October 2020, using SHRIMP-B. Analyses 1.1 to 6.1 (spot numbers 1–6) were obtained during the first session, together with 13 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.24% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.39%. Analyses 7.1 to 14.1 (spot numbers 7–14) were obtained during the second session, together with 16 analyses of the M257 standard, of which 14 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Analyses 10.2 to 8.2 (spot numbers 15–20) were obtained during the third session, together with 12 analyses of the M257 standard, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.31% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty analyses were obtained from 16 zircons. Results are listed in Table 1, and shown in concordia diagrams (Figs 3, 4).

Figure 1. Drillcore image (wet surface) of sample 203739: felsic metavolcanic rock, Lake Carey

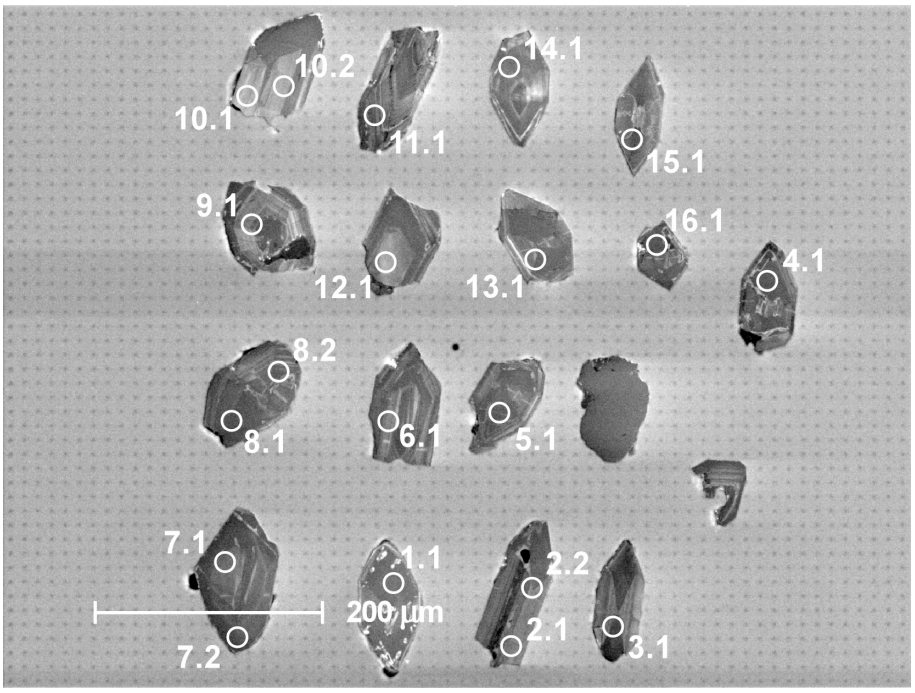

Figure 2. Cathodoluminescence image of all zircons from sample 203739: felsic metavolcanic rock, Lake Carey. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 203739: felsic metavolcanic rock, Lake Carey

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	20	8.2	262	129	0.49	-0.044	2.009	0.062	0.18164	0.00314	2.008	0.062	0.18203	0.00315	2606	67	2672	29	2.5
I	15	10.2	232	135	0.58	-0.025	2.014	0.123	0.18192	0.00138	2.013	0.123	0.18214	0.00140	2600	131	2673	13	2.7
I	14	14.1	343	170	0.49	0.048	1.968	0.023	0.18353	0.00109	1.969	0.023	0.18310	0.00112	2648	26	2681	10	1.2
I	10	10.1	123	50	0.40	0.000	1.945	0.032	0.18345	0.00177	1.945	0.032	0.18345	0.00177	2675	36	2684	16	0.4
I	19	7.2	304	132	0.43	0.080	1.966	0.040	0.18474	0.00110	1.967	0.040	0.18402	0.00114	2649	44	2690	10	1.5
I	13	13.1	268	152	0.57	0.066	1.948	0.025	0.18473	0.00129	1.950	0.025	0.18414	0.00134	2669	28	2691	12	0.8
I	17	16.1	405	191	0.47	0.012	1.965	0.021	0.18497	0.00096	1.966	0.021	0.18486	0.00097	2651	24	2697	9	1.7
I	2	2.1	147	55	0.37	0.028	1.947	0.027	0.18527	0.00162	1.947	0.027	0.18502	0.00164	2672	30	2698	15	1.0
I	8	8.1	467	272	0.58	0.000	1.972	0.030	0.18519	0.00093	1.972	0.030	0.18519	0.00093	2644	33	2700	8	2.1
I	4	4.1	457	226	0.49	0.048	1.983	0.020	0.18566	0.00115	1.984	0.020	0.18524	0.00117	2632	22	2700	10	2.5
I	1	1.1	411	233	0.57	0.041	1.953	0.020	0.18591	0.00117	1.954	0.020	0.18555	0.00119	2664	22	2703	11	1.4
I	5	5.1	387	210	0.54	-0.066	1.985	0.020	0.18502	0.00120	1.984	0.020	0.18562	0.00122	2631	23	2704	11	2.7
I	7	7.1	193	90	0.47	-0.027	1.970	0.027	0.18610	0.00142	1.969	0.027	0.18634	0.00144	2647	30	2710	13	2.3
I	18	2.2	170	68	0.40	-0.058	1.932	0.027	0.18658	0.00149	1.931	0.027	0.18710	0.00153	2690	32	2717	13	1.0
X	6	6.1	166	79	0.48	-0.155	2.000	0.026	0.18917	0.00160	1.997	0.026	0.19055	0.00169	2617	29	2747	15	4.7
X	12	12.1	96	35	0.37	-0.509	1.922	0.036	0.18664	0.00208	1.912	0.036	0.19118	0.00256	2712	42	2752	22	1.5
P	16	15.1	662	442	0.67	0.180	2.526	0.025	0.14055	0.00068	2.530	0.025	0.13896	0.00077	2147	18	2214	10	3.0
D	3	3.1	710	608	0.86	0.108	2.634	0.024	0.13760	0.00139	2.637	0.024	0.13665	0.00141	2073	17	2185	18	5.1
D	11	11.1	410	194	0.47	0.000	2.129	0.023	0.17913	0.00100	2.129	0.023	0.17913	0.00100	2483	23	2645	9	6.1
D	9	9.1	212	100	0.47	0.195	2.197	0.029	0.18737	0.00144	2.201	0.029	0.18562	0.00159	2414	27	2704	14	10.7

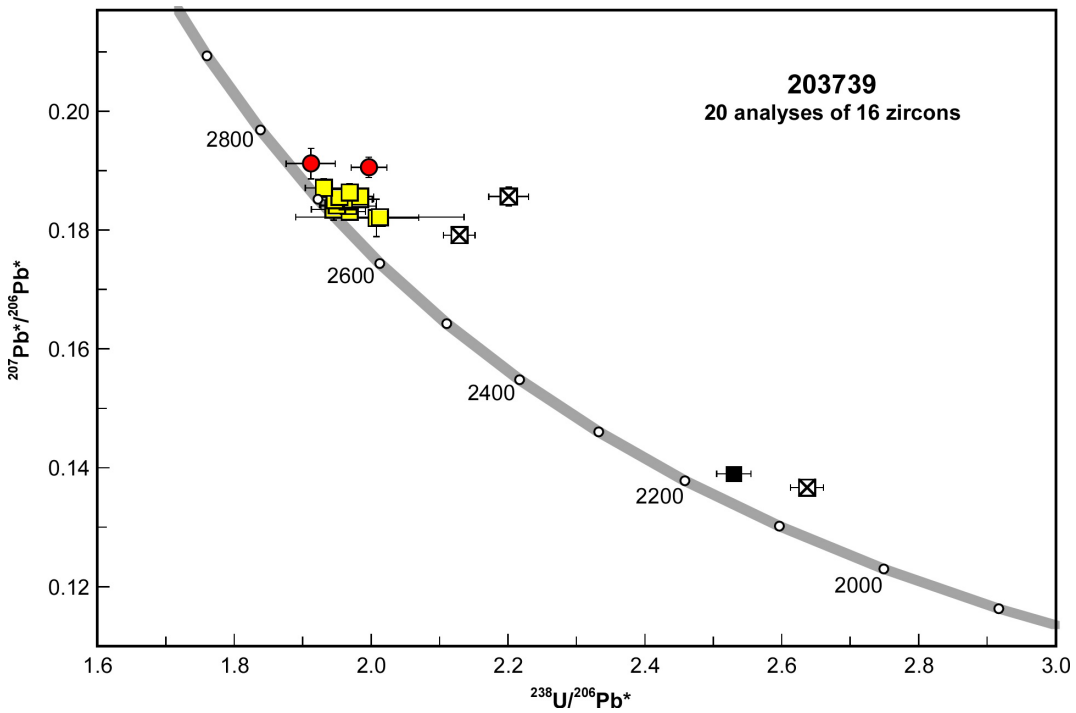

Figure 3. U–Pb analytical data for sample 203739: felsic metavolcanic rock, Lake Carey. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons); black square indicates Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%)

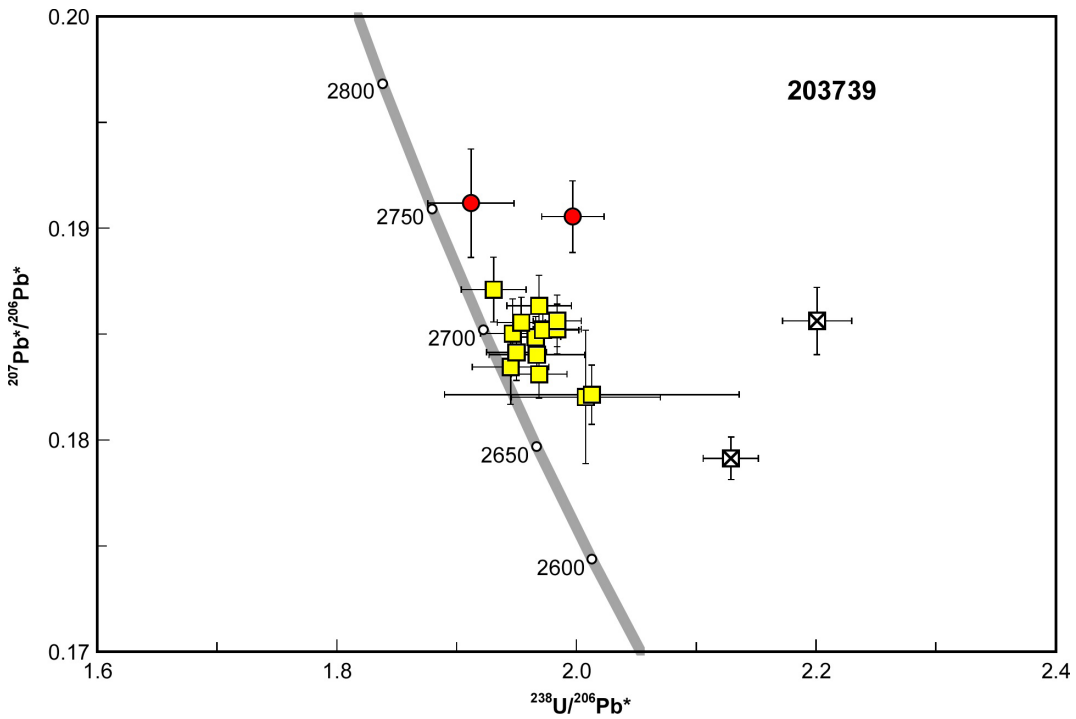

Figure 4. Expanded view of U–Pb analytical data for sample 203739: felsic metavolcanic rock, Lake Carey. Symbols as in Figure 3

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Three analyses are >5% discordant. The dates obtained from these three analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 17 analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 14 analyses of 10 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2696 ± 6 Ma (MSWD = 0.96).

Group X comprises two analyses of two zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2749 ± 24 Ma (MSWD = 0.041).

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2214 ± 10 Ma (1σ).

The date of 2696 ± 6 Ma for the 14 analyses in Group I is interpreted as age of igneous crystallization. The date of 2749 ± 24 Ma for the two analyses in Group X is interpreted as the age of an inherited component. The date of 2214 ± 10 Ma (1σ) for the single analysis in Group P is interpreted to reflect loss of radiogenic Pb.

References

Kositcin, N, Brown, SJA, Barley, ME, Krapež, B, Cassidy, KF and Champion, DC 2008, SHRIMP U-Pb zircon age constraints on the Late Archaean tectonostratigraphic architecture of the Eastern Goldfields Superterrane, Yilgarn Craton, Western Australia: Precambrian Research, v. 161, p. 5–33.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Standing, JG 2008, Terrane amalgamation in the Eastern Goldfields Superterrane, Yilgarn Craton: Evidence from tectonostratigraphic studies of the Laverton Greenstone Belt: Precambrian Research, v. 161, p. 114–134.

Stewart, AJ 2001, Laverton, WA Sheet SH 51-2; 1:250 000 Geological Series Explanatory Notes, 2nd Edition: Geological Survey of Western Australia, 37p

Wingate, MTD, Fielding, IOH, Lu, Y, Cassidy, KF and Smithies, RH 2022a, 203738: porphyritic metasyenite, Lake Carey; Geochronology Record 1881: Geological Survey of Western Australia, 4p.

Wingate, MTD, Fielding, IOH, Lu, Y, Cassidy, KF and Smithies, RH 2022b, 203740: porphyritic metasyenite, Wallaby mine; Geochronology Record 1883: Geological Survey of Western Australia, 4p.

Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y, Cassidy, KF and Smithies, RH 2022, 203739: felsic metavolcanic rock, Lake Carey; Geochronology Record 1882: Geological Survey of Western Australia, 5p.

Data obtained: 23 October 2020

Data released: 20 May 2022